



CYBERSECURITY LABORATORY REPORT

*Building and Validation of a
Cybersecurity Laboratory Environment*



KALI LINUX
Attacker Machine



DOCKER
Container Platform



OWASP JUICE SHOP
Vulnerable Web
Application



NMAP
Network Scanning
& Enumeration



BURP SUITE
Web Traffic
Interception



WIRESHARK
Network Traffic
Analysis

PREPARED BY

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OBJECTIVE:

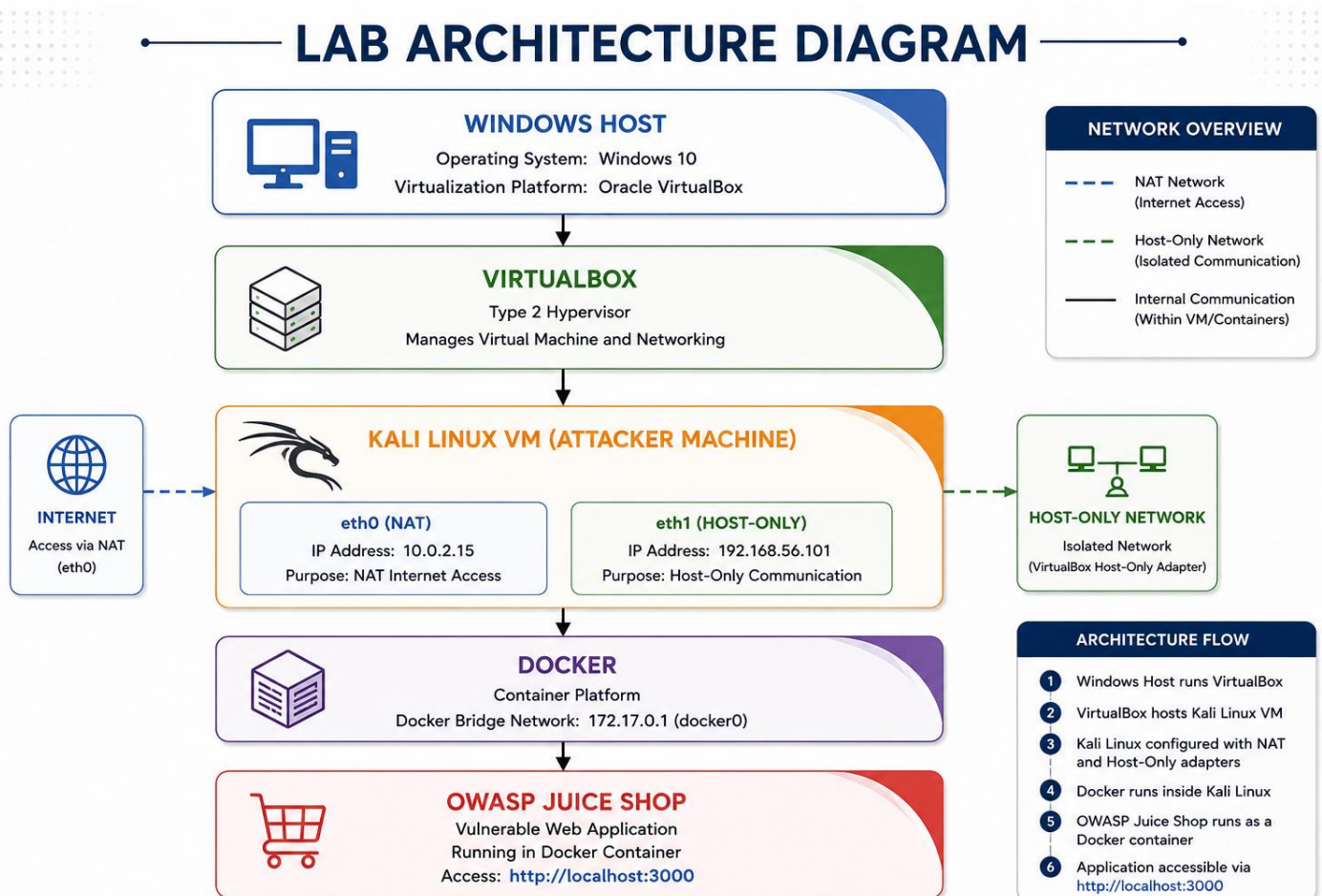
The goal of this task was to create a secure cyber security testing lab setup for penetration testing, vulnerability analysis, web app testing and network analysis. The lab is made up of an attacker machine running Kali Linux and a vulnerable target application of OWASP Juice Shop.

LAB ARCHITECTURE:

The lab environment was created using Oracle VirtualBox.

Components:

- Kali Linux (Attacker Machine)
- OWASP Juice Shop (Vulnerable Web Application)
- NAT Network Adapter
- Host-Only Network Adapter



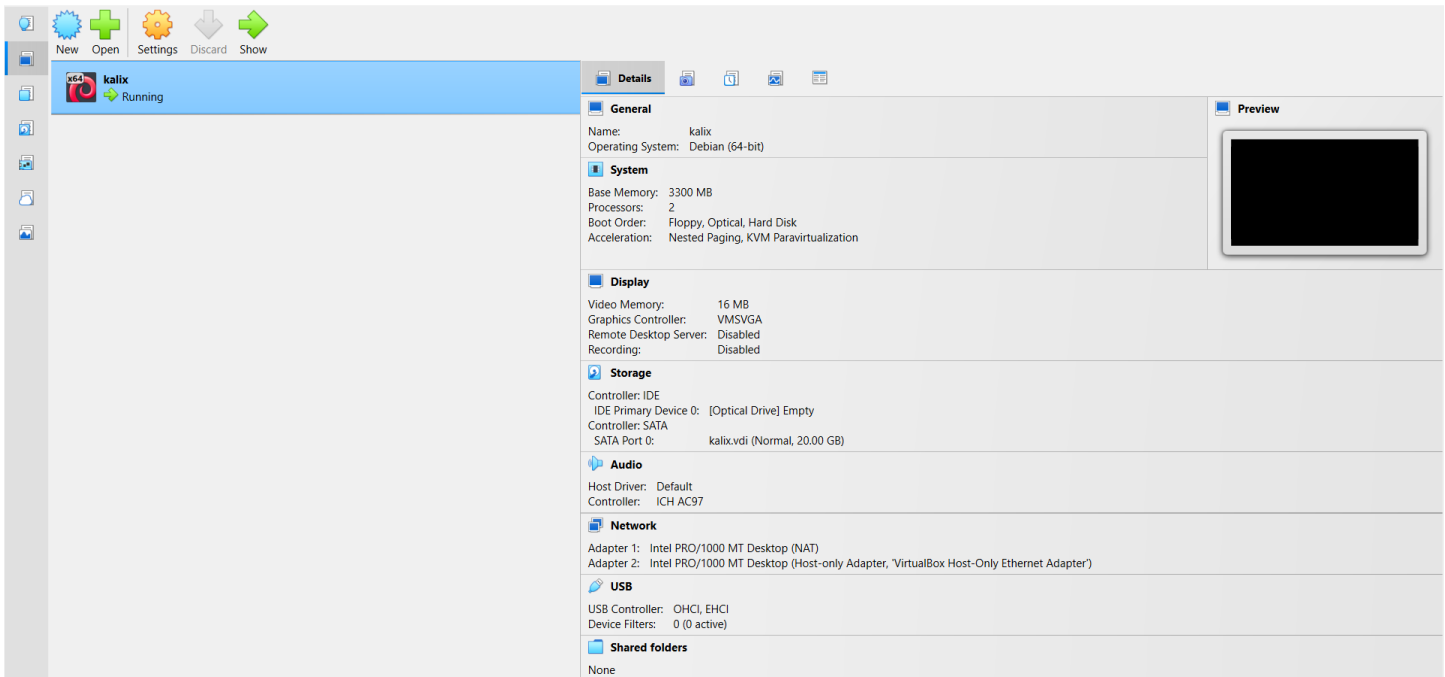


Figure 1: Oracle VirtualBox VM Configuration

SYSTEM REQUIREMENTS:

- Operating System: Windows 10
- RAM: 12 GB
- Virtualization: Enabled
- Virtualization Platform: Oracle VirtualBox
- Attacker Machine: Kali Linux
- Vulnerable Application: OWASP Juice Shop

VIRTUAL MACHINE CONFIGURATION:

Kali Linux was configured with:

- Adapter 1: NAT
- Adapter 2: Host-Only Adapter
- Internet connectivity enabled
- Host-only communication enabled

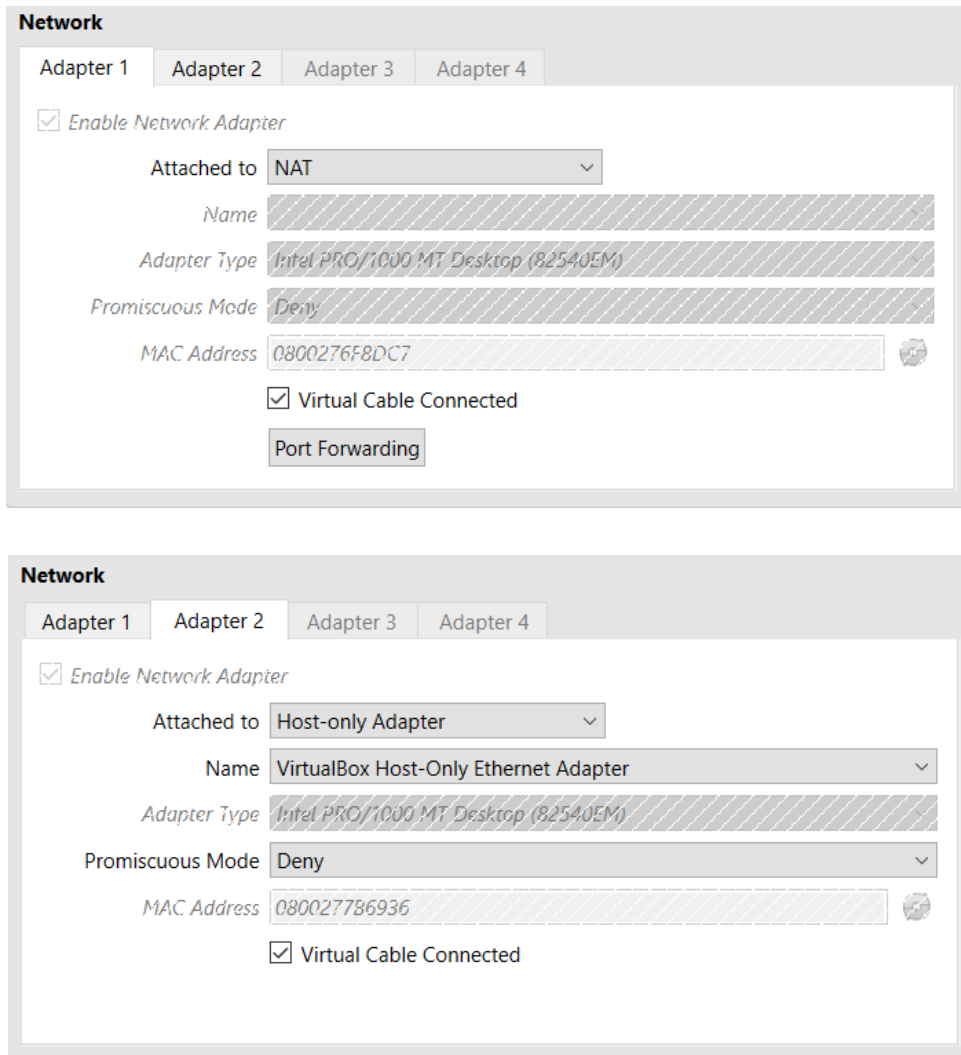


Figure 2: Virtual Network Adapter Configuration

NETWORK CONFIGURATION:

The network configuration was verified using the 'ip a' command.

Results:

- eth0 (NAT): 10.0.2.15
- eth1 (Host-Only): 192.168.56.101
- docker0 (Docker Bridge): 172.17.0.1


```

└─$ ip -4 addr show eth0
ip -4 addr show eth1
ip -4 addr show docker0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 85622sec preferred_lft 85622sec
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
       valid_lft 423sec preferred_lft 423sec
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
   inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever

```

Figure 3: IP Address Verification

Table 1: IP Addressing Information

Device	Interface	IP Address	Network Type	Purpose
Kali Linux VM	eth0	10.0.2.15/24	NAT Network	Internet Connectivity
Kali Linux VM	eth1	192.168.56.101/24	Host-Only Network	Host Communication
Docker Engine	docker0	172.17.0.1/16	Docker Bridge Network	Container Communication

CONNECTIVITY VALIDATION:

Internet connectivity was verified using ping commands.

Commands Used:

ping-c 4 8.8.8.8

ping-c 4 google.com

Results:

- Internet access confirmed
- DNS resolution confirmed

```

L$ ping -c 4 8.8.8.8
ping -c 4 google.com
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=4.65 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=5.38 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=4.77 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=4.84 ms

— 8.8.8.8 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 4.647/4.906/5.376/0.279 ms
PING google.com (142.251.221.206) 56(84) bytes of data.
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=1 ttl=64 time=19.2 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=2 ttl=64 time=4.65 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=3 ttl=64 time=4.49 ms
64 bytes from pnmaaa-ba-in-f14.1e100.net (142.251.221.206): icmp_seq=4 ttl=64 time=4.95 ms

— google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3013ms
rtt min/avg/max/mdev = 4.488/8.330/19.238/6.299 ms

```

Figure 4: Connectivity Validation Using Ping

OWASP JUICE SHOP DEPLOYMENT:

Docker was used to deploy OWASP Juice Shop.

Commands Used:

```
sudo docker pull bkimminich/juice-shop
```

```
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

The application was successfully accessed through:

<http://localhost:3000>

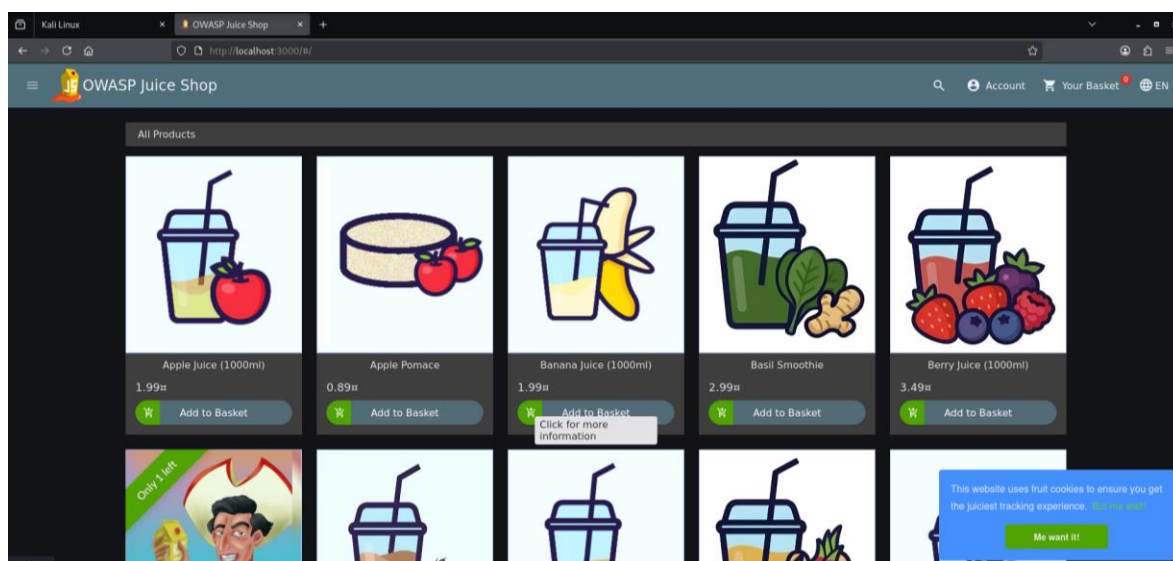


Figure 5: OWASP Juice Shop Deployment

NMAP VALIDATION:

Nmap (Network Mapper) is a network reconnaissance and security assessment tool which is used to identify open ports, to discover hosts and to detect services running in the target system. It provides security professionals with an understanding of the network exposure of systems and the ability to verify service availability.

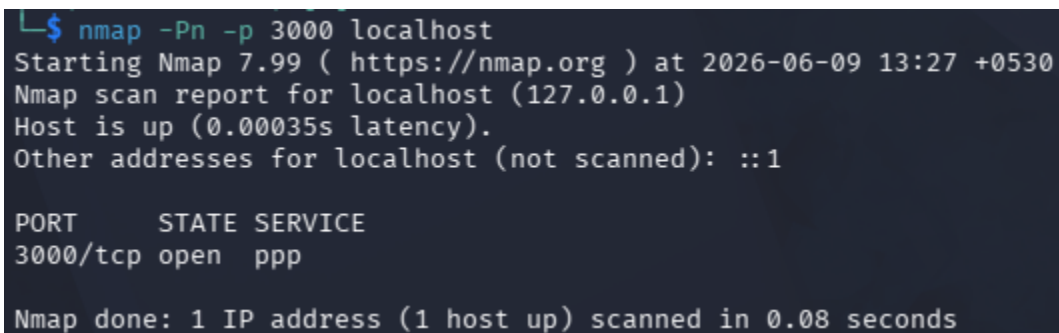
The Nmap tool was used in this lab to scan the local system and check if the application OWASP Juice Shop deployed via Docker is available. Services were detected on port 3000 to ensure that a service was listening and available.

Command used:

```
nmap -Pn -p 3000 localhost
```

Results:

- Port 3000/tcp identified as open
- Service version detection confirmed an active service listening on port 3000
- The service was verified to be OWASP Juice Shop through successful browser access to <http://localhost:3000>



```
└─$ nmap -Pn -p 3000 localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-09 13:27 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00035s latency).
Other addresses for localhost (not scanned): ::1

PORT      STATE SERVICE
3000/tcp  open  ppp

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
```

Figure 6: Nmap Service Validation

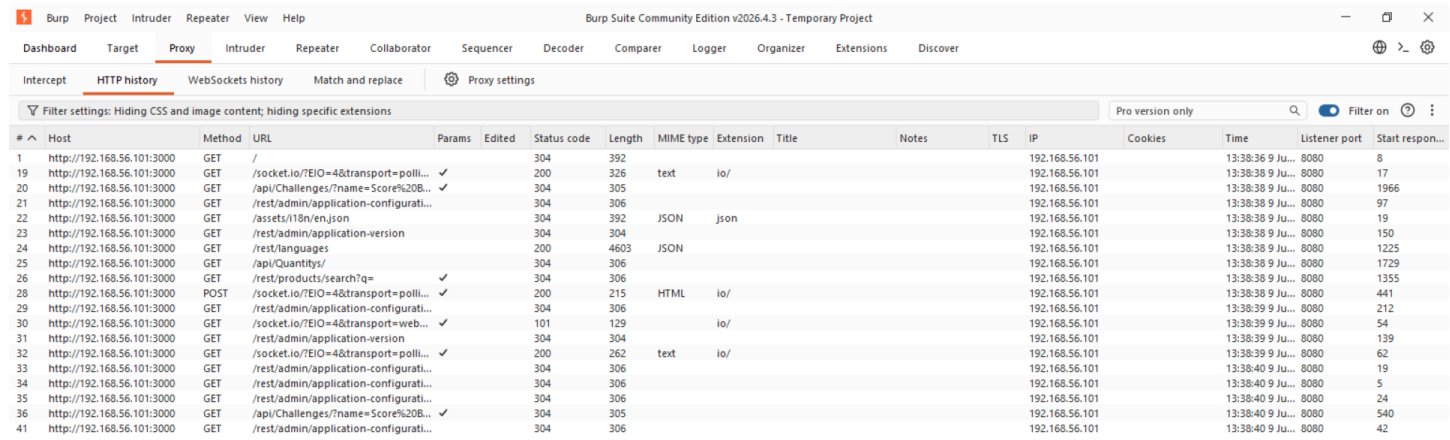
BURPSUITE VALIDATION:

Burp Suite Community Edition is a web application security testing tool designed to be used to intercept, inspect and analyze HTTP/HTTPS traffic between a web browser and a target application. It can help security analysts view web requests and responses, learn where vulnerabilities lie, and gain insight into how the application functions.

The set up for this lab was to run Burp Suite as a proxy between the browser and the OWASP Juice Shop application. All the HTTP traffic generated by the application was successfully captured and analyzed, validating the proper operation of traffic monitoring and analysis.

Results:

- Proxy successfully configured
- HTTP requests intercepted
- HTTP traffic successfully analysed and Proxy functionality successfully validated



The screenshot shows the Burp Suite interface with the 'HTTP history' tab selected. The table below represents the data visible in the history list.

#	Host	Method	URL	Params	Edited	Status code	Length	MIME type	Extension	Title	Notes	TLS	IP	Cookies	Time	Listener port	Start respon...
1	http://192.168.56.101:3000	GET	/			304	392						192.168.56.101		13:38:36 9 Ju...	8080	8
19	http://192.168.56.101:3000	GET	/socket.io/?EIO=4&transport=poll...		✓	200	326	text	io/				192.168.56.101		13:38:38 9 Ju...	8080	17
20	http://192.168.56.101:3000	GET	/api/Challenges/?name=Score%20B...		✓	304	305						192.168.56.101		13:38:38 9 Ju...	8080	1966
21	http://192.168.56.101:3000	GET	/rest/admin/application-configurati...			304	306						192.168.56.101		13:38:38 9 Ju...	8080	97
22	http://192.168.56.101:3000	GET	/assets/i18n/en.json			304	392	JSON	json				192.168.56.101		13:38:38 9 Ju...	8080	19
23	http://192.168.56.101:3000	GET	/rest/admin/application-version			304	304						192.168.56.101		13:38:38 9 Ju...	8080	150
24	http://192.168.56.101:3000	GET	/rest/languages			200	4603	JSON					192.168.56.101		13:38:38 9 Ju...	8080	1225
25	http://192.168.56.101:3000	GET	/api/Quantities/			304	306						192.168.56.101		13:38:38 9 Ju...	8080	1729
26	http://192.168.56.101:3000	GET	/rest/products/search?q=		✓	304	306						192.168.56.101		13:38:38 9 Ju...	8080	1355
28	http://192.168.56.101:3000	POST	/socket.io/?EIO=4&transport=poll...		✓	200	215	HTML	io/				192.168.56.101		13:38:38 9 Ju...	8080	441
29	http://192.168.56.101:3000	GET	/rest/admin/application-configurati...			304	306						192.168.56.101		13:38:39 9 Ju...	8080	212
30	http://192.168.56.101:3000	GET	/socket.io/?EIO=4&transport=web...		✓	101	129		io/				192.168.56.101		13:38:39 9 Ju...	8080	54
31	http://192.168.56.101:3000	GET	/rest/admin/application-version			304	304						192.168.56.101		13:38:39 9 Ju...	8080	139
32	http://192.168.56.101:3000	GET	/socket.io/?EIO=4&transport=poll...		✓	200	262	text	io/				192.168.56.101		13:38:39 9 Ju...	8080	62
33	http://192.168.56.101:3000	GET	/rest/admin/application-configurati...			304	306						192.168.56.101		13:38:40 9 Ju...	8080	19
34	http://192.168.56.101:3000	GET	/rest/admin/application-configurati...			304	306						192.168.56.101		13:38:40 9 Ju...	8080	5
35	http://192.168.56.101:3000	GET	/rest/admin/application-configurati...			304	306						192.168.56.101		13:38:40 9 Ju...	8080	24
36	http://192.168.56.101:3000	GET	/api/Challenges/?name=Score%20B...		✓	304	305						192.168.56.101		13:38:40 9 Ju...	8080	540
41	http://192.168.56.101:3000	GET	/rest/admin/application-configurati...			304	306						192.168.56.101		13:38:40 9 Ju...	8080	42

Figure 7: Burp Suite HTTP Interception

WIRESHARK VALIDATION:

Wireshark is a network protocol analyzer that can capture and display running network traffic. It enables cybersecurity experts to observe the interactions between devices, understand network protocols, troubleshoot connectivity problems, and delve into security events.

For this lab, ICMP traffic was captured and analyzed via Wireshark traffic, which confirmed successful network communication and packet analysis.

Result:

- Network interfaces successfully detected
- Live network traffic captured and analysed
- ICMP request and reply packets identified
- Packet analysis environment successfully validated

No.	Time	Source	Destination	Protocol	Length	Info
895	59.565121600	192.168.0.102	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=7/1792, ttl=128 (reply in 896)
896	59.568877100	8.8.8.8	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=7/1792, ttl=118 (request in 895)
904	60.574401800	192.168.0.102	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=8/2048, ttl=128 (reply in 905)
905	60.591740300	8.8.8.8	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=8/2048, ttl=118 (request in 904)
907	61.577828600	192.168.0.102	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=9/2304, ttl=128 (reply in 908)
908	61.581473800	8.8.8.8	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=9/2304, ttl=118 (request in 907)
909	62.582259400	192.168.0.102	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=10/2560, ttl=128 (reply in 910)
910	62.585707600	8.8.8.8	192.168.0.102	ICMP	74	Echo (ping) reply id=0x0001, seq=10/2560, ttl=118 (request in 909)

Frame 895: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF...		0000	f0 a7 31 f8 66 ae 80 01 33 d6 ad f1 08 00 45 00	..1 f... 3... E-
> Ethernet II, Src: AzureWaveTec_d6:ad:f1 (80:91:33:d6:ad:f1), Dst: TPLink_f8:66:ae (f0:a7:31:f8:66:ae)	0010	00 3c 73 cc 00 00 80 01 f5 d6 c0 a8 00 66 08 08	<S.....f..	
> Internet Protocol Version 4, Src: 192.168.0.102, Dst: 8.8.8.8	0020	08 08 08 00 4d 54 00 01 00 07 61 62 63 64 65 66	...MT...abcdef	
> Internet Control Message Protocol	0030	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijklmn opqrstuv	
	0040	77 61 62 63 64 65 66 67 68 69	wabcedfg hi	

Figure 8: Wireshark Packet Capture

KEY OUTCOMES:

- Successfully deployed Kali Linux in Oracle VirtualBox.
- Installed NAT and Host-Only network adapters.
- Verified network connectivity and DNS resolution.
- Successfully deployed OWASP Juice Shop using Docker.
- Set up an application that is accessible via port 3000.
- Used Nmap to obtain a list of services.
- Enumerated services with Nmap.
- Intercepted HTTP traffic and succeeded (with Burp Suite)
- Used Wireshark to capture and analyze network traffic.
- Set up a functional Cybersecurity Lab Environment.

CONCLUSION:

The cyber security lab environment has been designed, deployed and validated successfully with the help of Oracle VirtualBox, Kali Linux, Docker and OWASP Juice Shop. The network connectivity and service availability were checked using several tools such as Ping, Nmap, Burp Suite and Wireshark. Final product environment: Completely secure and isolated space for cybersecurity training, web application security testing, vulnerability assessment, and network traffic analysis.

REFERENCES:

1. OWASP Juice Shop Documentation
2. Kali Linux Documentation
3. Docker Documentation
4. VirtualBox User Manual
5. Nmap Reference Guide
6. PortSwigger Burp Suite Documentation
7. Wireshark User Guide
8. OWASP Web Security Testing Guide (WSTG)